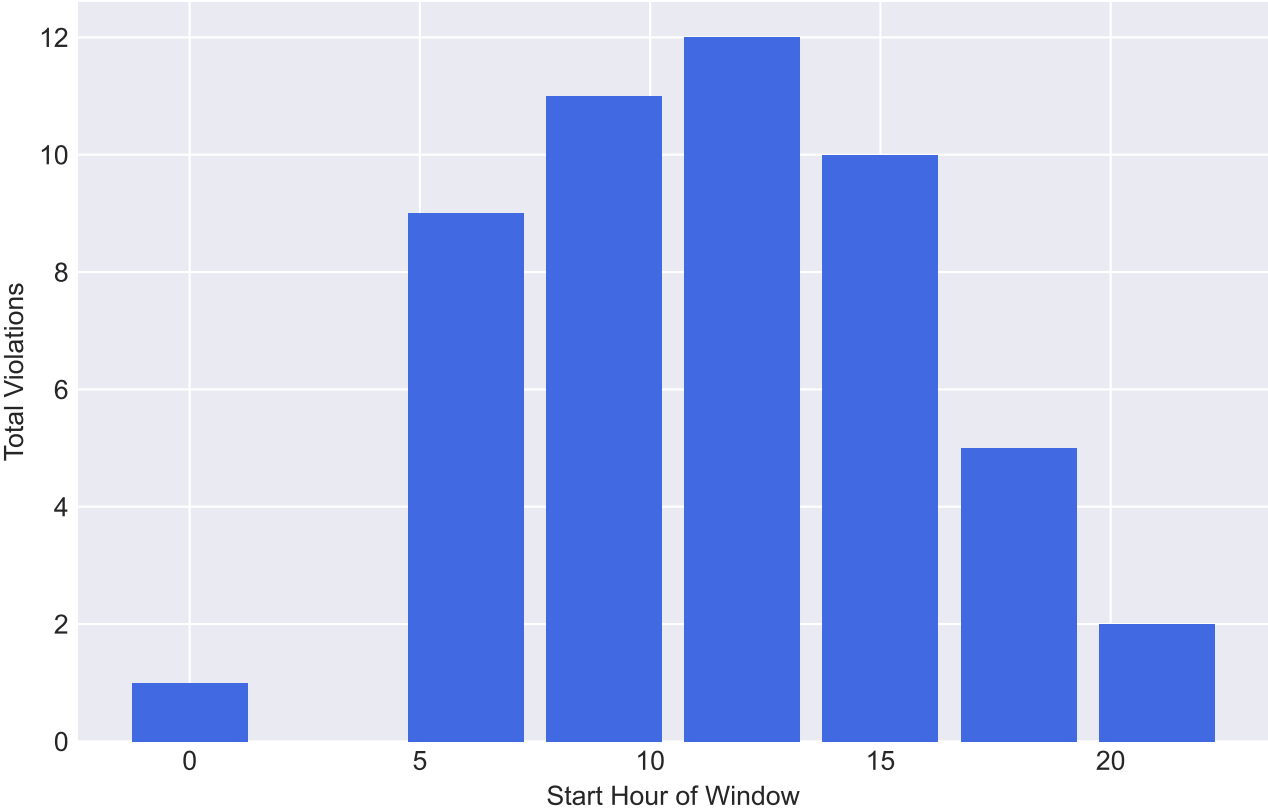
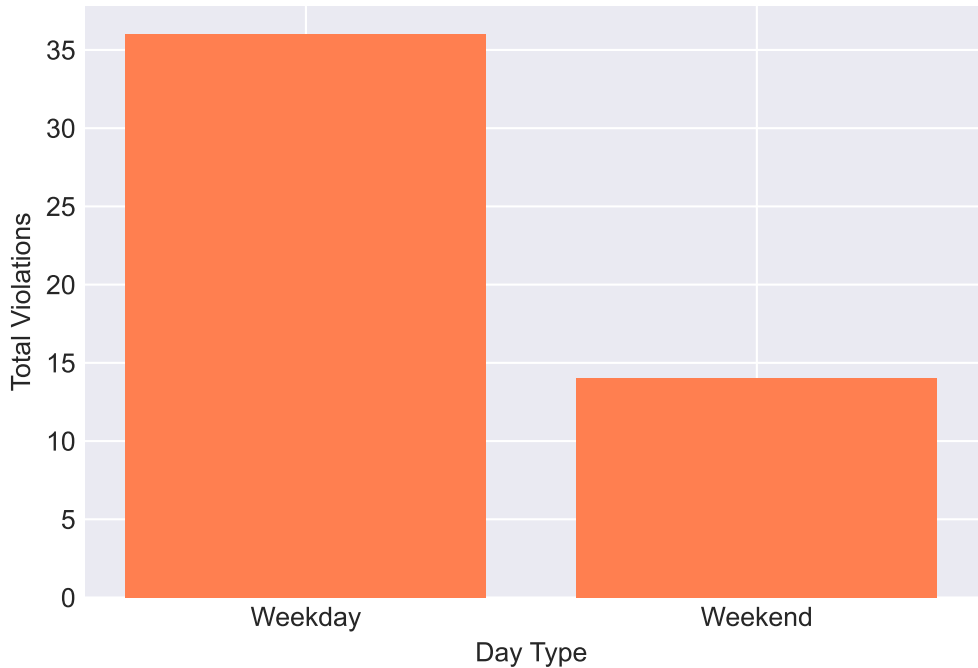


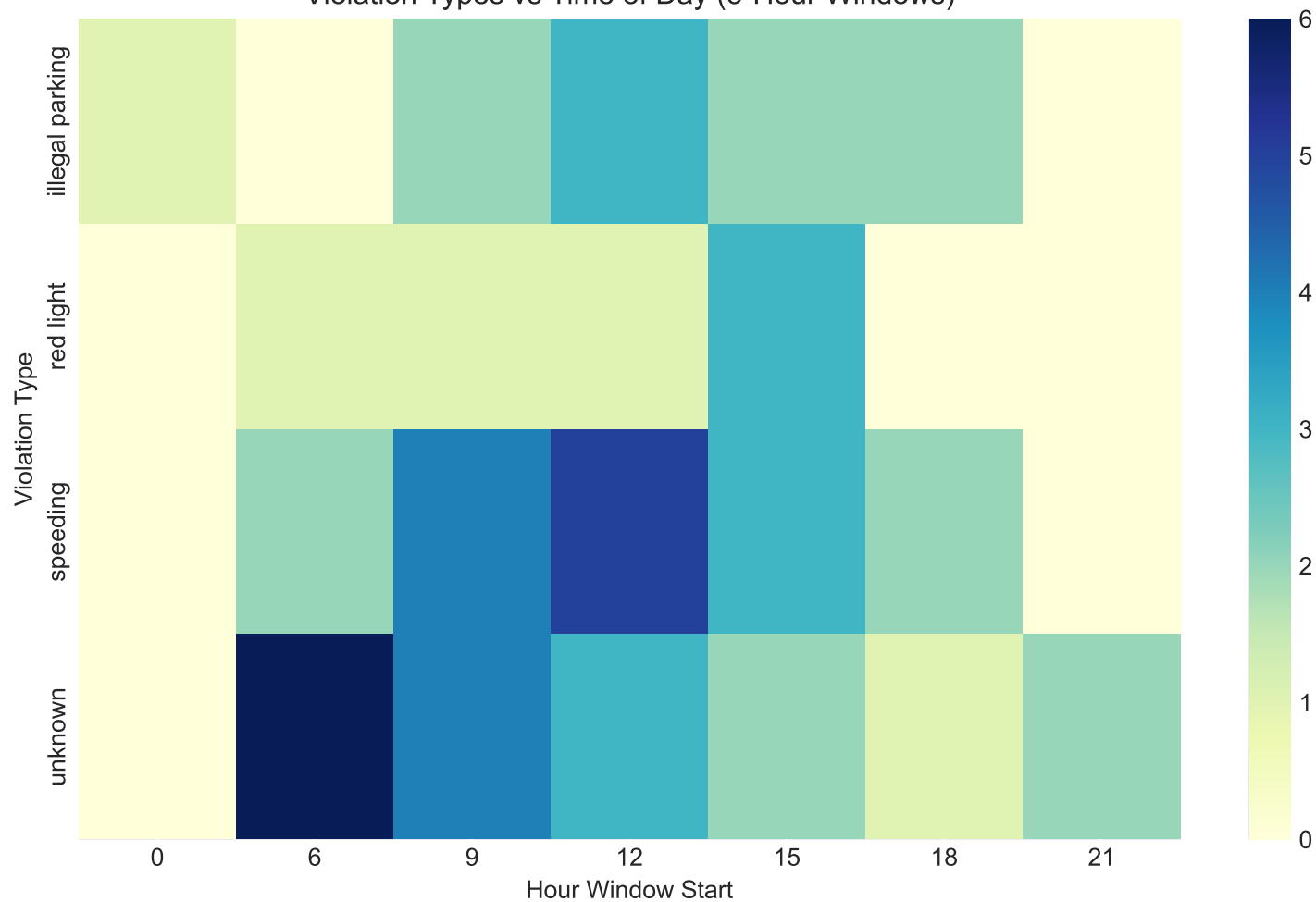
Violations per 3-Hour Window



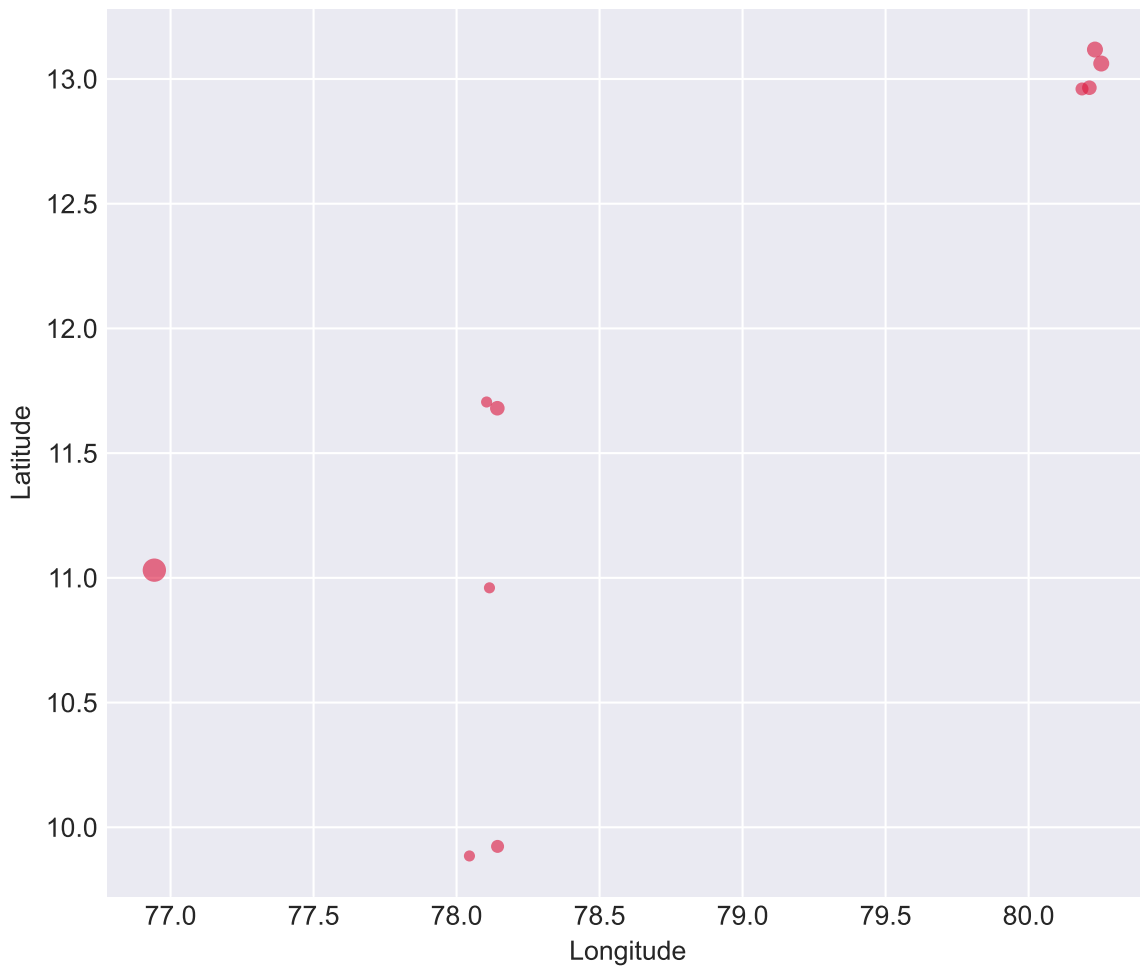
Violations: Weekday vs Weekend



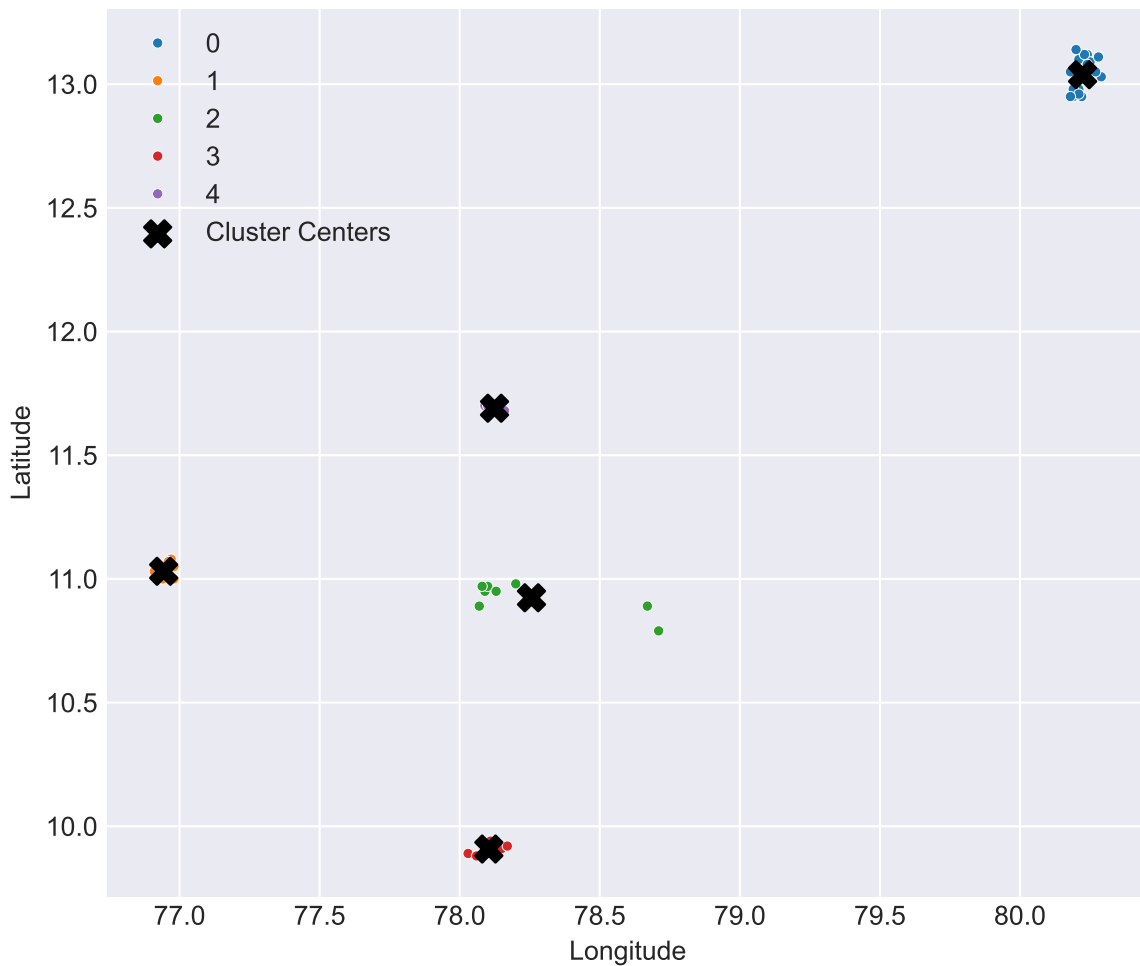
Violation Types vs Time of Day (3-Hour Windows)



Top 10 Hotspot Grid Cells



K-Means Violation Hotspot Clusters (5 Zones)



□ ADVANCED TRAFFIC VIOLATION ANALYSIS

Records analyzed: 50

Key Highlights:

- Time grouping in 3-hour windows
- Weekday vs Weekend comparison
- Violation-type vs Time heatmap
- Spatial grid hotspot detection
- (Optional) K-Means clustering on coordinates

Top Hotspot Zones:

grid_cell	total_violations	avg_lat	avg_lon
110_769	12	11.030833	76.943333
131_802	5	13.118000	80.232000
130_802	5	13.062000	80.254000
116_781	4	11.680000	78.142500
129_802	4	12.965000	80.212500